

## COLLARDRIVE · DESIGN DECISION RECORD

# CollarDrive TRL 2 review decisions

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## DOCUMENT CONTROL

|             |                                                            |         |                |
|-------------|------------------------------------------------------------|---------|----------------|
| Document ID | CLD-DDR-001                                                | Version | 0.1            |
| Status      | Draft                                                      | Date    | 2026-10-03     |
| Author      | Amish Chadha                                               | License | CERN-OHL-S-2.0 |
| Project TRL | TRL 3 Proof of concept                                     |         |                |
| Repository  | github.com/BoujeeEnjinia1701/collardrive                   |         |                |
| Source      | docs/decisions/0001-trl2-review-decisions.md @ uncommitted |         |                |

## REVISION HISTORY

| Version | Date       | Author       | Change                                                                     |
|---------|------------|--------------|----------------------------------------------------------------------------|
| 0.1     | 2026-10-03 | Amish Chadha | TRL 2 review items D1 to D14 decided under Amish's 2026-10-03 pre-approval |

# 0001: TRL 2 review decisions

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- **Date:** 2026-10-03
- **Status:** accepted. Decided by Amish under his pre-approval of 2026-10-03: "start with the first 14 repos from the list of 29 projects. I pre-approve the batch runs along with any recommendations you come up with. I also accept any cost overruns or variations from the assumed scope cost."

## ■ Context

The TRL 2 work for CollarDrive (precis CLD-PRC-001 v0.2, requirements CLD-REQ-001 v0.2) left the choices below to be made before the TRL 3 calculations and model. Amish pre-approved every recommendation for this batch, so each is recorded as decided, with the recommendation as the decision. Choices that touch safety take the conservative option and say what evidence would relax them. Partners and regions are the first candidates to approach, not agreements.

**Safety:** CollarDrive works next to an open shaft with a running engine and belts. Its purpose is to keep exhaust out of the shaft; it does not make a shaft safe to enter. Gas testing, collar fencing and stopping the engine before touching the drive are part of every decision below.

## ■ Options considered

*Table 1. Options for each item.*

| #   | Item                            | Options                                                                                                                                                                    |
|-----|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| D1  | Engine                          | (a) 4.8 kW (6.5 hp) class petrol engine with a horizontal shaft; (b) small diesel; (c) electric motor where grid or solar exists                                           |
| D2  | Drive layout                    | (a) Engine to a layshaft, then separate belts to the pump and the blower; (b) engine directly to the pump with the blower on the engine's other shaft end; (c) two engines |
| D3  | Pump disconnect                 | (a) Belt-tensioning idler clutch on the pump belts; (b) bought centrifugal or dog clutch; (c) take a belt off by hand                                                      |
| D4  | Rope pump                       | (a) Rope and washer pump with a tyre-sidewall wheel and a rising main sized by depth; (b) one pipe size for all depths                                                     |
| D5  | Ventilation                     | (a) Forcing (blowing) through 200 mm lay-flat duct; (b) exhausting through rigid duct                                                                                      |
| D6  | Exhaust                         | (a) Pipe to a riser 6 m or more downwind; (b) short tail pipe and rely on wind                                                                                             |
| D7  | Frame and ground                | (a) Two bolted modules on three staked timber sleepers set back from the collar; (b) one welded frame bearing on the collar                                                |
| D8  | Running back of the rope        | (a) Ratchet backstop on the wheel shaft; (b) none                                                                                                                          |
| D9  | Guarding                        | (a) Full belt guard, layshaft cover, wheel hood and blower inlet guard; (b) partial guards                                                                                 |
| D10 | Gas detector                    | (a) Personal CO and oxygen detector in the kit; (b) left to the user                                                                                                       |
| D11 | Scope                           | (a) Pump and air only, no hoisting, frame never stood on; (b) add an ore hoist                                                                                             |
| D12 | Design shaft for the priced kit | (a) 30 m with 30 m of duct; (b) 60 m                                                                                                                                       |
| D13 | Co-design partners              | First candidates to approach                                                                                                                                               |
| D14 | First trial region              | First candidate to approach                                                                                                                                                |

## ■ Decision

Table 2. Decisions.

| #   | Decision                                                                                                                                                                                                                                                                                                                           | Reason                                                                  | Status                                      |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|---------------------------------------------|
| D1  | (a) 4.8 kW class 196 cc petrol engine, horizontal 19.05 mm keyed shaft, governed to 2,800 rpm; a diesel of similar size fits the same plate                                                                                                                                                                                        | Sold and repaired everywhere; five times the power needed               | Decided by Amish, 2026-10-03 (pre-approval) |
| D2  | (a) Engine to a layshaft with one A belt; layshaft to the pump with two matched A belts and to the blower with one A belt                                                                                                                                                                                                          | One engine, slow wheel and fast blower from standard pulleys            | Decided by Amish, 2026-10-03 (pre-approval) |
| D3  | (a) Over-centre idler clutch on the slack strand of the pump belts, lever latched in the belt guard                                                                                                                                                                                                                                | No bought clutch; air can run alone                                     | Decided by Amish, 2026-10-03 (pre-approval) |
| D4  | (a) Rope pump, 400 mm rope circle; 40 mm PVC main to 30 m, 32 mm to 45 m, 25 mm to 60 m                                                                                                                                                                                                                                            | Keeps the rope pull inside the belts' capacity at every depth           | Decided by Amish, 2026-10-03 (pre-approval) |
| D5  | (a) Forcing ventilation through 200 mm lay-flat duct, blower intake facing the front of the frame                                                                                                                                                                                                                                  | Fresh air delivered to the face; light duct that hangs from rings       | Decided by Amish, 2026-10-03 (pre-approval) |
| D6  | (a) 40 mm pipe on stands to a riser with a rain cap, outlet 2.4 m high and at least 6 m from the collar edge and the blower intake, set downwind of the prevailing wind. Conservative choice: the 6 m minimum stays until carbon monoxide readings at the blower intake over a range of wind directions show a shorter run is safe | Follows the NIOSH 6 m (20 ft) advice                                    | Decided by Amish, 2026-10-03 (pre-approval) |
| D7  | (a) Span module over the opening and drive module beyond it, bolted, on three sleepers whose edges stay at least 0.35 m back from the collar edge; stakes through the sleepers                                                                                                                                                     | Nothing bears on the collar lip; each piece can be carried              | Decided by Amish, 2026-10-03 (pre-approval) |
| D8  | (a) Ratchet backstop on the pump shaft. Conservative choice: kept even for shallow shafts, until a test shows the wheel cannot run back hard enough to hurt                                                                                                                                                                        | The water column runs the rope back when the engine stops               | Decided by Amish, 2026-10-03 (pre-approval) |
| D9  | (a) Full guarding of every belt, pulley, shaft and the wheel; the blower inlet meshed                                                                                                                                                                                                                                              | R9; miners work close to the drive                                      | Decided by Amish, 2026-10-03 (pre-approval) |
| D10 | (a) A personal carbon monoxide and oxygen detector with alarm is part of the kit (BOM line 29), used before and during work below ground. Conservative choice: required in every kit; nothing relaxes it                                                                                                                           | Ventilation alone does not make a shaft safe                            | Decided by Amish, 2026-10-03 (pre-approval) |
| D11 | (a) Pump and air only; no hoist; the frame is never stood on and the collar is fenced (fence supplied by the site)                                                                                                                                                                                                                 | Hoisting people or ore is a different and much higher hazard            | Decided by Amish, 2026-10-03 (pre-approval) |
| D12 | (a) The priced kit is for a 30 m shaft with 30 m of duct; deeper shafts change the main and add duct                                                                                                                                                                                                                               | Middle of the 10 to 60 m range                                          | Decided by Amish, 2026-10-03 (pre-approval) |
| D13 | First candidates to approach: an artisanal mining cooperative through a national artisanal miners' body, with a university mining department; and a local rope pump maker, through the Practica Foundation's rope pump network. Not agreed                                                                                         | Real collars and depths; existing pump makers who can build and service | Decided by Amish, 2026-10-03 (pre-approval) |

| #   | Decision                                                                                                                                                | Reason                                                              | Status                                      |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|---------------------------------------------|
| D14 | First candidate region to approach: Zimbabwe (Mashonaland West, where engine fumes have killed artisanal miners), then Nigeria (Kano State). Not agreed | Documented deaths from engines in shafts; large artisanal workforce | Decided by Amish, 2026-10-03 (pre-approval) |

## ■ Consequences

- The TRL 3 calculations (CLD-CAL-001), the model and the general arrangement (CLD-DWG-001) use these decisions.
- The concept media, the precis and the requirements are updated to match.
- Making the design constructable led to further changes, recorded in CLD-DDR-002.
- Partners and the region are recorded as first candidates; contacting them is Amish's step.